Linear Regression

Machine Learning 1 Practicals

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March 18, 2026

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- Inputs: features $x_1, x_2, \ldots, x_D$, collected in $\mathbf{x} = (x_1, x_2, \ldots, x_D)^T \in \mathbb{R}^D$
- Given $\mathbf{x}$, we want to predict a **real-valued target** $y \in \mathbb{R}$

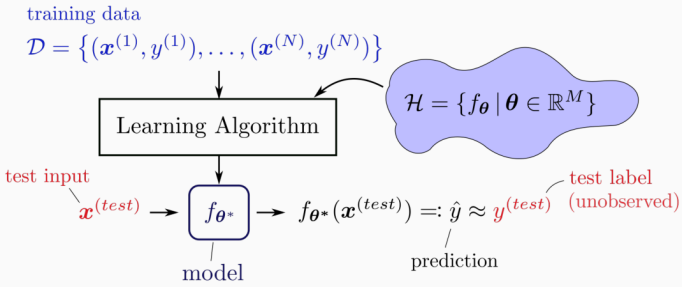
Example

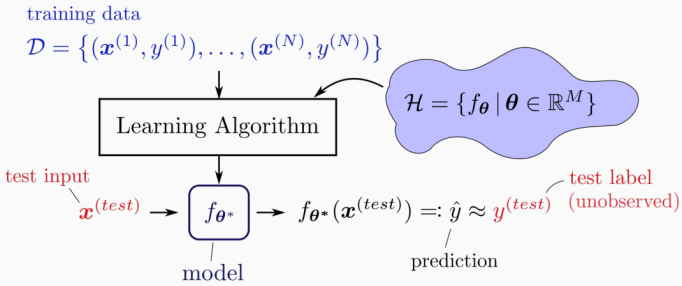
Given knee height in cm (x_1) and arm span in cm (x_2),
predict the body height in cm (y).

- We collect a **training dataset** with many such input-output pairs

$$\mathcal{D} = \left\{ \left(\mathbf{x}^{(1)}, y^{(1)} \right), \ldots, \left(\mathbf{x}^{(N)}, y^{(N)} \right) \right\}$$

Recap: Supervised Learning





- **Linear Regression:** We assume an (approx.) **affine relationship** between $\mathbf{x}$ and y:

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = b + w_1 \cdot x_1 + w_2 \cdot x_2 + \cdots + w_D \cdot x_D$$

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- Collect $\boldsymbol{\theta} = (b, w_1, w_2, \ldots, w_D)^T$
- Let's add a **constant feature** (dummy feature) to each $\mathbf{x}$:

$$\mathbf{x} = (1, x_1, \ldots, x_D)^T$$

- We can now compactly write

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\theta}$$

Loss function ❗

- We need a way to **quantify how "good" a given $\boldsymbol{\theta}$** is
- We call this the **loss** $\mathcal{L}(\boldsymbol{\theta})$
  - Small values are "good"

- In regression, we often use the **Mean Squared Error** as the loss:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \left(f_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)} \right)^2$$

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- Collect data points in the **rows** of a matrix (**data matrix** X) and targets in a **vector**

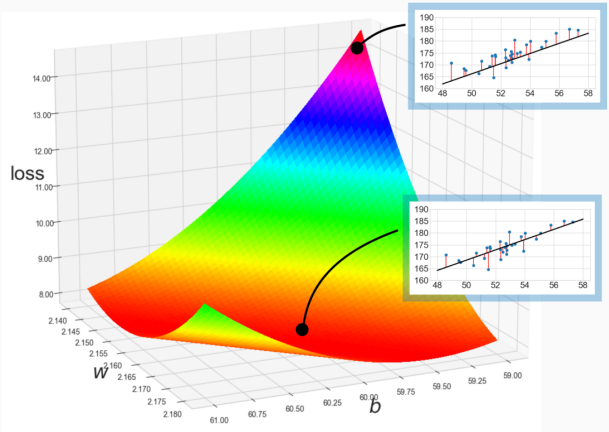
$$X = \begin{pmatrix} \mathbf{x}^{(1)^T} \\ \mathbf{x}^{(2)^T} \\ \vdots \\ \mathbf{x}^{(N)^T} \end{pmatrix} \qquad \mathbf{y} = \left(y^{(1)}, y^{(2)}, \ldots, y^{(N)}\right)^T \in \mathbb{R}^n$$

- Then, we can re-write the loss as

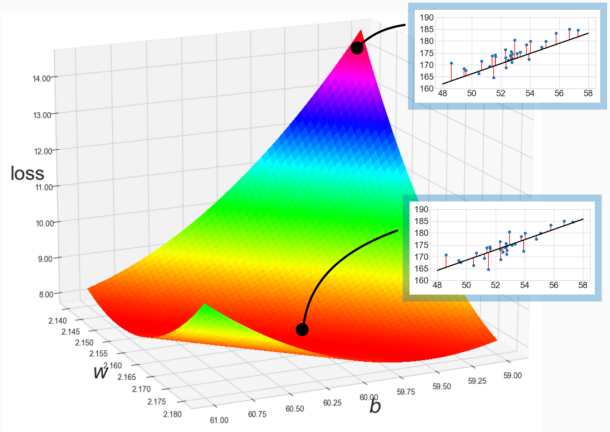
$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N}\|X\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

Minimizing the loss function

This choice of $\mathcal{L}$ and f_θ yields a loss function that is **convex in θ** 🥰



This choice of $\mathcal{L}$ and $f_{\boldsymbol{\theta}}$ yields a loss function that is **convex in $\boldsymbol{\theta}$** 🥰



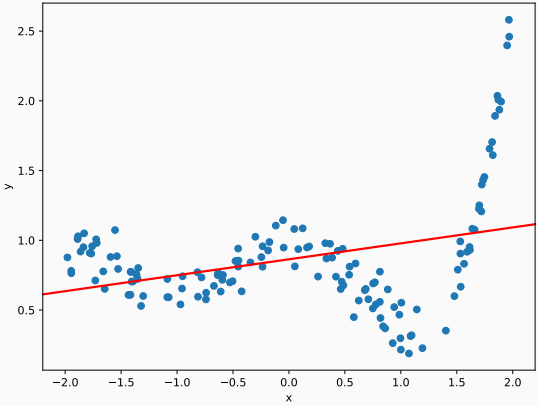
- **Goal:** find

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\, \mathcal{L}(\boldsymbol{\theta})$$

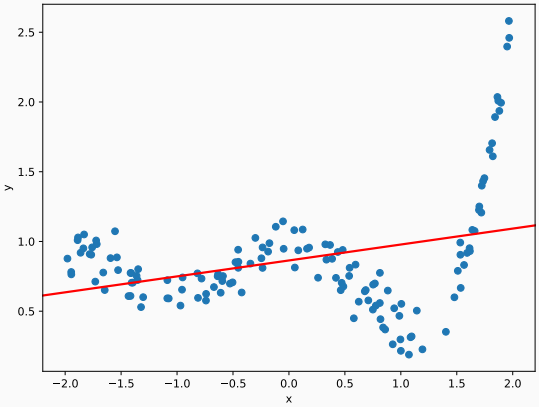
- $\boldsymbol{\theta}^*$ is **unique** iff X has **full rank**
- **There exists a closed-form solution for the global minimum** 😍

$$\boldsymbol{\theta}^* = \underbrace{(X^T X)^{-1} X^T}_{\text{Pseudoinverse}} \mathbf{y}$$

Linear-in-the-Parameters Regression

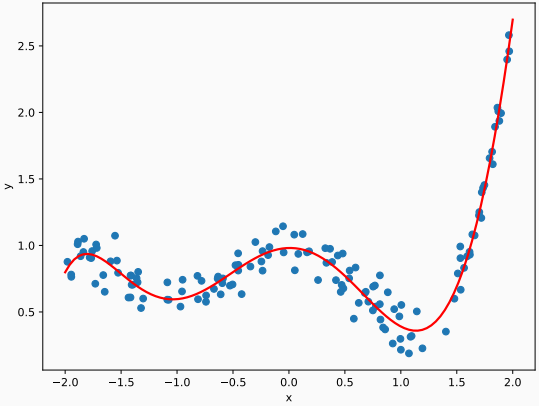


Is this the result of **linear** regression? 🤔

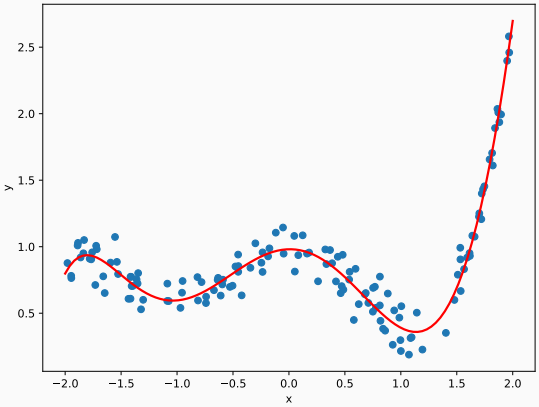


Yes! The function in red is of the form

$$f_{\boldsymbol{\theta}}(x) = b + w \cdot x$$



Is this still **linear** regression? 🤔



Polynomial Regression is still **linear** in $\boldsymbol{\theta}$ **!**

$$f_{\boldsymbol{\theta}}(x) = b + w_1 \cdot x + w_2 \cdot x^2 + \cdots + w_6 \cdot x^6$$

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- It's just **non-linear** in x
- Let $\phi(x) = (1 \;\; x \;\; x^2 \;\; x^3 \ldots \;\; x^6)^T$

$$f_{\boldsymbol{\theta}}(x) = \phi(x)^T \boldsymbol{\theta}$$

Dimensionality

- **Note:** x is 1D while $\phi(x)$ and $\boldsymbol{\theta}$ are 7D ❗
- We lift x into a higher dimensional space
- Hypothesis: There is an affine relationship between the **transformed features** $\phi(x)$ and the target

Linear Regression Demo